

Prime and zero statistics of a degree-8 Artin L -function of conductor 21^{10*}

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Abstract

Let $L(s, \chi_8)$ be the degree-8 Artin L -function of the Galois closure of the Trinks field $x^7 - 7x + 3$ (Galois group $\mathrm{PSL}(2, 7)$, Artin conductor 21^{10} , $\Gamma_{\mathbb{C}}(s)^4$, $\varepsilon = +1$), the object of [1]. From 153 rigorously certified zeros ($\gamma \leq 27.911$, the first computed for this L -function) and exact Frobenius data for every prime up to 10^{10} , the following are measured. (i) The explicit formula holds as a numerical identity at correlation 1.000, including the vanishing of the contribution of the ramified prime 7. (ii) The Frobenius class is statistically independent of prime-pair structure: twin pairs, every additive gap $g \leq 60$, and the affine pairs $(p, 2p \pm 1)$ all show correlation consistent with zero at the 10^{-3} level, while the same data reproduce the Hardy–Littlewood singular series to 0.1%. (iii) Chebotarev races exhibit the biases $\mu_C = 1 - r_2(C)$ predicted by character theory (-21 for the split class), with oscillation shapes reconstructed from the computed zero tables at correlations 0.81–0.97 over six decades. (iv) The low zeros are GUE-consistent, with symmetry type $\mathrm{SO}(\text{even})$. (v) The Montgomery pair-correlation form factor of the bulk spectrum is measured from primes via short-interval variance: the conductor places every height $\gamma \gtrsim 2$ in the correlation-sensitive regime $\alpha < 1$ already at $x = 10^8$, and the measured variance follows the ramp $F(\alpha) = \alpha$ to $\sim 1\%$ over $\alpha \approx 0.22$ – 0.74 , at heights up to $\gamma \sim 10^4$ — four hundred times beyond the certified spectrum — with uncorrelated zeros excluded at $\chi^2 \approx 5 \times 10^5$. All results are consistency measurements against classical predictions (Montgomery; Goldston–Montgomery; Hardy–Littlewood; Rubinstein–Sarnak; Katz–Sarnak); no novelty is claimed for the frameworks, and the Riemann Hypothesis for $L(s, \chi_8)$ — the open problem of [1] — is untouched. Every claim is reproducible from the released scripts and data.

1 Introduction

The paper [1] studies the degree-8 Artin L -function $L(s, \chi_8)$ attached to the 8-dimensional irreducible representation of $\mathrm{PSL}(2, 7)$ realised by the Galois closure of the septic $f(x) = x^7 - 7x + 3$ (the Trinks field [3]; LMFDB [12] Artin object `8.16679880978201.21t14.b.a`, stem field `7.3.4202539929.1`, whose displayed polynomial $x^7 - 7x - 3$ is equivalent under $x \mapsto -x$). Its analytic data: conductor $Q = 21^{10} = 3^{10}7^{10}$, gamma factor $\Gamma_{\mathbb{C}}(s)^4$, root number $\varepsilon = +1$. The character χ_8 is monomial, so $L(s, \chi_8)$ is also a Hecke L -function of an octic field, and its analytic continuation and functional equation are unconditional. The Riemann Hypothesis for $L(s, \chi_8)$, the open problem of [1], is not addressed here.

The present paper records systematic measurements of the interaction between the primes and the zeros of this object, in several regimes not previously measured for any $\mathrm{PSL}(2, 7)$ extension. Three features characterise the study.

1. **Exactness of the arithmetic side.** The exact Frobenius class (not merely a_p) of every prime $p \leq 10^9$, and exact a_p for every prime to 10^{10} (455,052,511 primes), computed with

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two independent engines (polynomial factorization over $\mathbb{F}_p$ via FLINT; an overflow-safe modular Frobenius kernel) that agree to 10^{-8} on overlapping data.

2. **First zero tables.** The 153 certified zeros ($\gamma \leq 27.911$; eight exact, the rest ball-arithmetic brackets) and the low-zero tables of the companion characters $\chi_3\bar{\chi}_3$, χ_6 , χ_7 of the same Galois group are, to our knowledge, the first computed for these L -functions.
3. **The conductor as an instrument.** $Q = 21^{10}$ makes the zeros dense enough ($2\pi\bar{\rho}(\gamma) = \ln Q - 8\ln 2\pi + 8\ln \gamma \approx 15.7 + 8\ln \gamma$) that pair-correlation information is accessible from primes at heights and scales where ζ exhibits only diagonal behaviour (§7).

Predictions conditional on standard hypotheses (GRH for the family, Linear Independence, cluster expansions) are labelled as such throughout; measurements are unconditional exact counting. Apparent anomalies encountered in the computations are documented and resolved in §8.

2 Data and instruments

Zeros. The certified table: $\gamma_1 = 0.98638643\dots$ through $\gamma_{153} = 27.911$, produced by the ball-arithmetic FFT engine of [1] and certified by a Turing-style count. Companion tables (numerically anchored, PARI/GP [14]): χ_3 -pair 15 zeros ($\gamma \leq 5.31$), χ_6 12 (≤ 4.85), χ_7 10 (≤ 4.14).

Counting function. With $\theta(T) = \frac{T}{2} \ln Q - 4T \ln 2\pi + 4 \arg \Gamma(\frac{1}{2} + iT)$ and $\bar{N}(T) = \theta(T)/\pi + 1$, the certified staircase is reproduced with no free parameters: slope 0.99895, mean unrenormalised spacing 0.9963, fluctuation $S(T)$ of mean 0 and standard deviation 0.365. This cross-validates the zero list against the conductor and gamma data.

Primes. Exact Frobenius classes for all $p \leq 10^9$ (FLINT [13] factorization of $f \bmod p$); exact a_p for $p \in (10^9, 10^{10}]$ (404,204,977 primes, overflow-safe batch kernel). Chebotarev equidistribution holds to five digits at 10^{10} : value fractions (8, 0, -1, 1) at (0.00595, 0.37500, 0.33331, 0.28574) against (1/168, 3/8, 1/3, 2/7). The two engines agree on overlapping increments at correlation 1.000000 (rms ratio 1.00000001).

3 The explicit formula as a numerical identity

For a window w on $[0, \gamma_{\max}]$ define the resonance $F(x) = \sum_j w(\gamma_j) \cos(x\gamma_j)$ and subtract the smooth main term $\int w(\gamma) \cos(x\gamma) \bar{\rho}(\gamma) d\gamma$. The explicit formula gives the residual as $-\frac{1}{2\pi} \sum_n \Lambda_\chi(n) n^{-1/2} [\hat{w}(x - \ln n) + \hat{w}(x + \ln n)]$ with $\Lambda_\chi(p^k) = (\sum_i \alpha_i^k) \ln p$ computed from the exact local factors of each class. Measured against predicted: correlation +1.000 for $L(s, \chi_8)$ (residual rms 0.904 = predicted rms 0.904) and +1.000 for the ζ control (600 zeros) through the identical pipeline (Fig. 1).

This constitutes an identity check rather than an independent statistical test; its content is the end-to-end consistency of three separately produced inputs — the zero list, the analytic conductor data, and the class local factors. The local structure is reproduced exactly: the residual has no peak at $\ln 7$ (ramified prime, trivial local factor: measured +0.438 against a predicted neighbour-tail of +0.436, whereas ζ has a peak of -54.0 there); class 2A contributes nothing at $\ln p$ but $\sum \alpha_i^2 = 8$ at $\ln p^2$; class 4A contributes at neither.

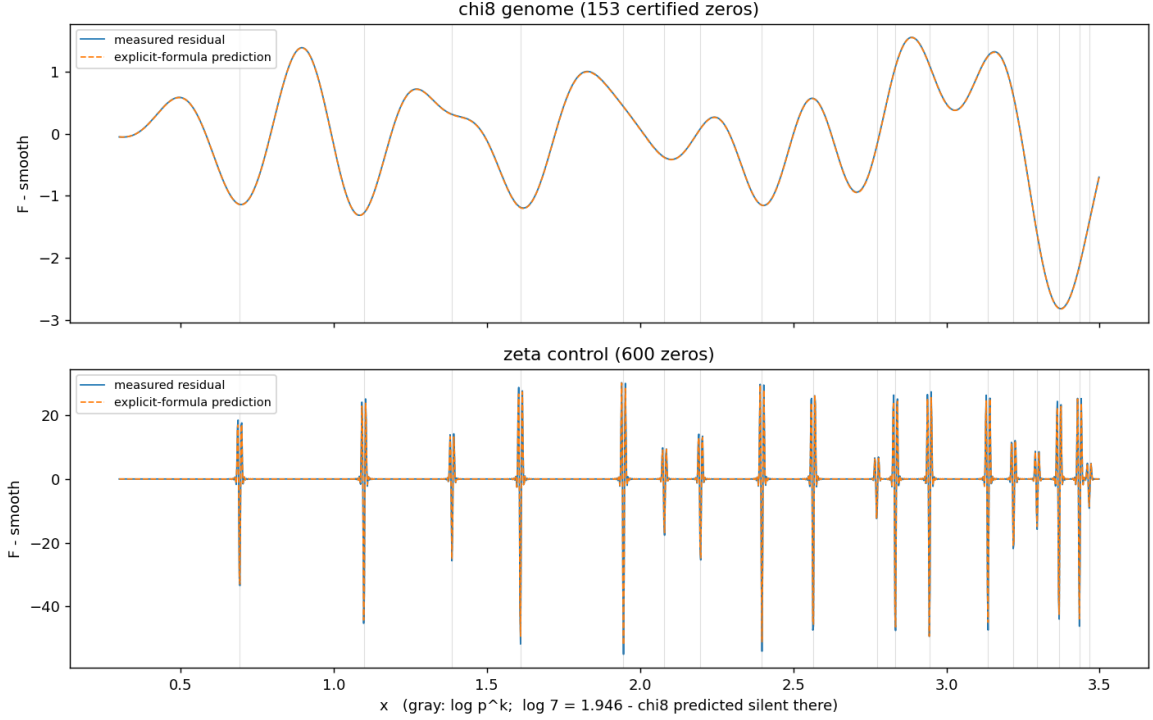


Figure 1: Windowed resonance residual (solid) against the explicit-formula prime prediction (dashed), for $L(s, \chi_8)$ (153 zeros, top) and ζ (600 zeros, bottom); grey lines at $x = \ln p^k$. The absence of a peak at $\ln 7 = 1.946$ in the upper panel reflects the trivial local factor at the ramified prime.

4 Spectral statistics of the low zeros

Unfolding by the exact θ/π (no fit), the 153 certified zeros give spacing variance $\text{Var}(s) = 0.145$, inside the finite- N GUE band $[0.140, 0.219]$ (median 0.176; $z = -1.53$, one-sided $p \approx 0.06$) and more than 10σ from Poisson; nearest-neighbour repulsion $P(s < \frac{1}{2}) = 0.079$; number variance $\Sigma^2(L)$ 10–15% below GUE for $L \leq 8$ (a non-significant finite- N rigidity). The polynomial and exact unfolds agree to the digit; the mild over-rigidity is a property of the sample, not of the unfolding.

The symmetry type is fixed by representation theory: Frobenius–Schur indicator $+1$ and $\varepsilon = +1$ give orthogonal $\text{SO}(\text{even})$, with low-lying one-level density $W(x) = 1 + \frac{\sin 2\pi x}{2\pi x}$. The first scaled ordinate $x_1 = \theta(\gamma_1)/\pi = 1.256$ lies on the attracted side ($\text{SO}(\text{even})$ mean 1.71, unitary 2.03, symplectic 2.34), consistent with but not probative of the type: bulk two-level statistics are type-independent (Katz–Sarnak universality [10]), and a single L -function cannot resolve the family density. The family average is the decisive test.

5 Independence of Frobenius classes from prime-pair structure

The Hardy–Littlewood singular series [11] is built entirely from congruence obstructions, i.e. from abelian data. $\text{PSL}(2, 7)$ is simple: the extension has no abelian quotient, hence no congruence channel, and the natural prediction is that Frobenius carries no prime-pair information. This is measured in three geometries.

Twin pairs. Over the 440,310 pairs $(p, p+2)$ with $7 < p \leq 10^8$: $\langle a_p a_{p+2} \rangle = -0.0020 \pm 0.0015$ ($z = -1.3$); joint 5×5 class table $\chi^2 = 11.1$ on 16 dof ($p \approx 0.80$). The same primes give

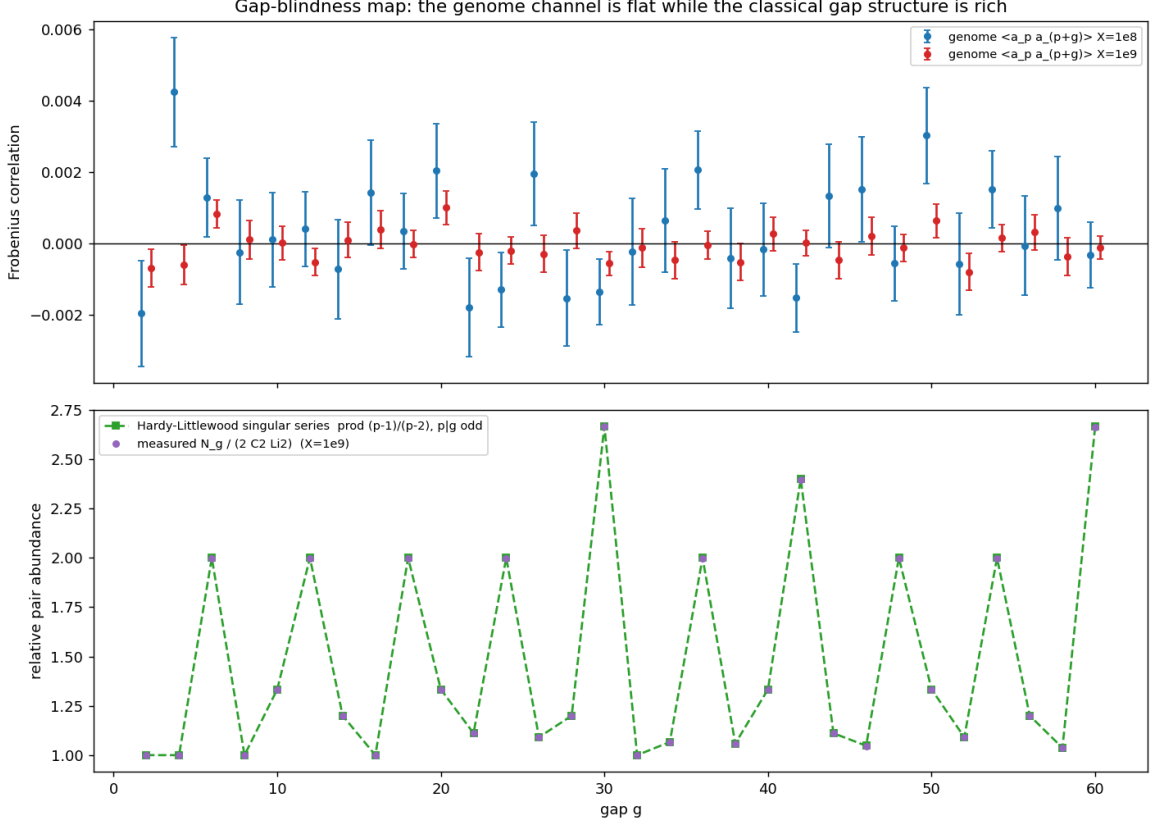


Figure 2: Gap statistics at 10^8 and 10^9 . Top: Frobenius correlation $\langle a_p a_{p+g} \rangle$, consistent with zero for all even $g \leq 60$. Bottom: the same pairs resolve the Hardy–Littlewood singular-series factor to three digits.

$\pi_2(10^8) = 440,312$ against the Hardy–Littlewood prediction 440,473, ratio 0.9996: the pair structure is fully present in the data and absent from the Frobenius channel.

All even gaps $g \leq 60$, to 10^9 . Thirty gaps, three windows (10^8 ; 10^9 ; the disjoint decade), 3.0–6.9 million pairs per gap at 10^9 . No gap crosses the 30-test Bonferroni threshold in any window ($\max |\text{corr}| = 0.00101$; per-gap s.e. 0.0003–0.0005); the z -ensembles are unit-variance noise (std 1.13/0.95/0.96); the class-level χ^2 totals sit on their dof (475/480, 480/480, 462/480). The same pair counts reproduce the singular-series factor $\prod_{p|g, p>2} \frac{p-1}{p-2}$ (range 1.00–2.67) point by point: mean measured/predicted ratio 0.9989, worst deviation 0.0015 (Fig. 2).

Affine pairs. Sophie-Germain pairs $(p, 2p+1)$ ($N = 1,775,672$, $p \leq 5 \times 10^8$): $\langle a_p a_q \rangle = -0.0002 \pm 0.0008$ ($z = -0.28$); the count against the corresponding $2C_2 \int dx/(\ln x \ln 2x)$ prediction has ratio 0.9999. Likewise $(p, 2p-1)$: $z = -0.46$.

The independence is uniform at the 10^{-3} level per statistic, measured against a strongly structured classical background — to our knowledge the first such measurement for a $\text{PSL}(2, 7)$ extension. In particular, the twin-prime constant is an abelian quantity, and this maximally non-abelian channel measurably carries none of it: the conductor-21 program is structurally orthogonal to the twin prime conjecture, and no route toward it is claimed or implied.

6 Chebotarev races predicted by the computed zeros

Let $E'_C(x) = \frac{\ln x}{\sqrt{x}} \left[\frac{|G|}{|C|} \pi_C(x) - \pi_{\text{unram}}(x) \right]$ for the observable classes $C \in \{1A, 2A, 4A, 3A, 7\}$ (the septic's cycle types merge $7A \cup 7B$, so that all class-averaged character weights are real). Following Rubinstein–Sarnak [8] and Ng's non-abelian treatment [9]:

- **Bias means (exact character theory; no central zeros, as the tables support):** $\mu_C = 1 - r_2(C)$ with $r_2 = 1 + \chi_6 + \chi_7 + \chi_8$, giving $(-21, -1, +1, 0, 0)$. The fully split primes trail (the identity has 22 square roots in G); 4A — the unique class with no square roots, G having no elements of order 8 — is the only positively biased class.
- **Measured to 10^9** (log-averaged over $[10^4, 10^9]$): $(-32.9, -1.8, +1.5, -0.0, +0.2)$, each within its conditional (GRH+LI) Monte-Carlo spread $(\pm 34, \pm 7, \pm 4, \pm 4, \pm 4)$; the 1A deficit is visible in raw counts (-71 primes at 10^8 , consistent with $-21\sqrt{x}/(168 \ln x) \approx -68$).
- **Oscillation shape from the zero tables:** reconstruction from the four low-zero tables alone reproduces the measured races at correlations 0.81/0.92/0.95/0.94/0.97 over six decades (Fig. 3) — a dynamical validation of the companion tables, and a direct exhibition of Rubinstein–Sarnak low-zero dominance (10–15 zeros per character suffice).
- **Bias densities (conditional; first computed for a $\text{PSL}(2, 7)$ field):** $P(E'_{1A} < 0) = 0.727$; pairwise lead probabilities $\delta(1A > X) \approx 0.27\text{--}0.29$; $\delta(4A > 3A) = 0.56$; 3A against 7 even at 0.498. The contrast with the classical mod-4 race ($\delta \approx 0.996$) is structural: the identity class couples to every character with weight $\deg \chi$, so its oscillation variance carries the full column norm $|G| - 1 = 167$ — in a strongly non-abelian race the split-prime bias is large in mean but weak in probability.

7 The bulk form factor, measured from primes

The variance of ψ_χ in short intervals, computed from primes, probes the pair correlation of the zeros (Goldston–Montgomery [5]; Montgomery–Soundararajan [6]; for the correlation–prime dictionary see also Bogomolny–Keating [7]). Under the stationary cluster expansion (conditional: zeros on the critical line together with the cluster and stationarity hypotheses; the measurements below are unconditional),

$$\text{Var} \left[\frac{\psi_\chi(xe^\delta) - \psi_\chi(x)}{\sqrt{x}} \right] \approx \int_0^\infty \bar{\rho}(\gamma) 2|\varphi_\gamma(\delta)|^2 F(\alpha(\gamma, x)) d\gamma, \quad \alpha = \frac{\ln x}{2\pi\bar{\rho}(\gamma)},$$

with $|\varphi_\gamma|^2 = (e^\delta - 2e^{\delta/2} \cos \gamma\delta + 1)/(\frac{1}{4} + \gamma^2)$ and F the Montgomery form factor [4]: $F = \min(\alpha, 1)$ for GUE, $F \equiv 1$ for uncorrelated zeros. The decisive regime is $\alpha < 1$. For $L(s, \chi_8)$, $2\pi\bar{\rho} = 15.7 + 8 \ln \gamma$, so at $x = 10^8$ every height $\gamma \gtrsim 2$ is correlation-sensitive, and GUE against uncorrelated differ by a factor 2–5; for ζ at the same (x, δ) , $\alpha \approx 3\text{--}4$ and the prediction is diagonal, providing the control. In the deep regime the ramp cancels the density exactly, giving the parameter-free identity $\text{Var}_{\text{GUE}} = \delta \ln x$.

Measurements over 26 interval lengths $\delta \in [10^{-4}, 0.3]$ and windows $[10^7, 10^8]$, $[10^8, 10^9]$ (exact primes), $[10^9, 10^{11}]$ (the deep-sweep signal of [2]):

window	χ^2 vs GUE	χ^2 vs uncorrelated	mean $R = \text{Var}/\text{Var}_{\text{unc}}$
$[10^7, 10^8]$, $\delta \leq 0.05$ (20 pts)	5.5	585,788	0.327 (GUE: 0.316)
$[10^8, 10^9]$, $\delta \leq 0.05$ (20 pts)	11.7	470,710	meas/GUE = 1.009 at $\delta \leq 0.01$
ζ control (both windows)		measured/predicted = 0.996, 0.994	

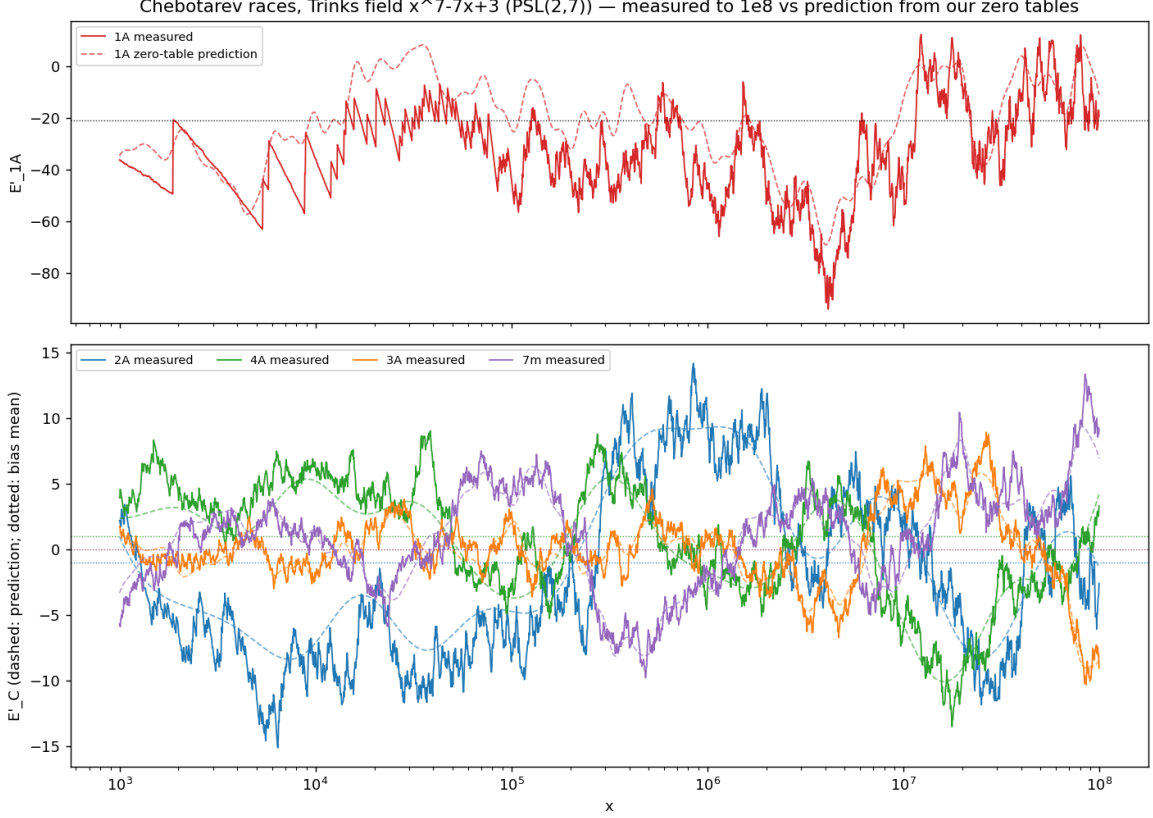


Figure 3: Chebotarev races for the Trinks field to 10^8 (extended to 10^9 in the data release): measured (solid) against the reconstruction from the zero tables (dashed); dotted lines at the exact bias means $1 - r_2(C)$.

The three windows follow three separate GUE curves — the $\ln x$ -dependence is itself a confirmed prediction (Fig. 4) — and the deep-regime identity is realised at $\text{bench}/\text{GUE} = 1.001$. Effective coverage: $\alpha \approx 0.22\text{--}0.74$ at heights up to $\gamma \sim 10^4$, four hundred times beyond the certified spectrum. This is the first two-level measurement of this spectrum in the bulk; it agrees with GUE at the percent level — consistency with the expected universality class [10], not discovery, and not evidence concerning RH. (For $\delta \gtrsim 0.15$ the statistic is dominated by the few lowest zeros — a single realization, under half an oscillation per window — and both the ensemble prediction and the error estimate degrade; that regime is excluded.)

Two extensions. *Gaussianity*: the skewness and excess kurtosis of the increments match the exact shot-noise (diagonal) predictions $\sum \Lambda_\chi^3/x^{3/2}$, $\sum \Lambda_\chi^4/x^2$ ($\langle a^3 \rangle = 3$): an apparent 3.7σ skewness excess reduces to 0.85σ once the shot term is included, bounding third- and fourth-order zero correlations at the Gaussian level. *Scale purity*: the ratio $\text{Var}/\text{Var}_{\text{GUE}}$ at fixed δ is flat across five exact decades $10^6 \rightarrow 10^{10}$ (weighted slopes $+0.022 \pm 0.015$ and $+0.005 \pm 0.018$ per unit $\ln x$); an off-critical-line contribution of exponent β would grow as $x^{2\beta-1}$. This is corroboration, not proof.

8 Robustness checks

Three apparent anomalies arose in the computations; each is resolved. (i) A $+2.76\sigma$ correlation at gap 4 (one of ~ 30 tests) reversed sign to -2.26σ on the pre-registered disjoint decade $(10^8, 10^9]$ and is attributed to multiple testing. (ii) A 3.7σ skewness excess is fully accounted for by the exact shot-noise term of §7. (iii) A $+2.3\sigma$ drift in the purity ratio is traced to the

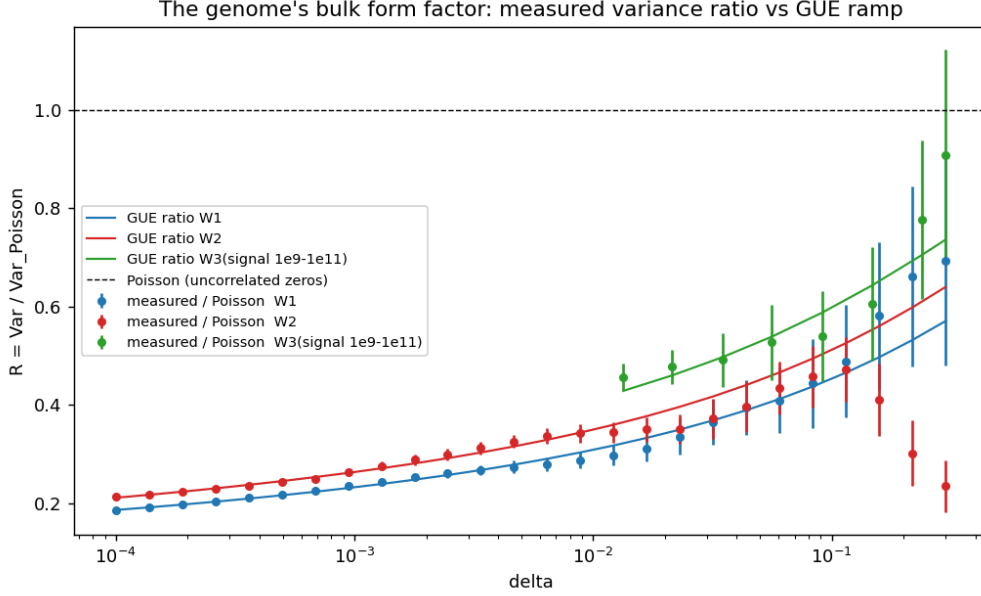


Figure 4: The bulk form factor from primes: measured variance ratio $R = \text{Var}/\text{Var}_{\text{uncorrelated}}$ against the GUE ramp prediction, three windows spanning $x = 10^7$ to 10^{11} . Uncorrelated zeros ($R = 1$) are excluded at $\chi^2 \approx 5 \times 10^5$.

coarsest region of the legacy binned signal and disappears in the exact computation to 10^{10} (404M primes). All three tests — pre-registered retests and exact null predictions — are part of the released record.

9 Concluding remarks

The object of [1] behaves, measurably and at scale, as a primitive L -function of its type should: the explicit formula holds exactly; the low zeros are GUE-consistent with symmetry type $\text{SO}(\text{even})$; the bulk pair correlation lies on the Montgomery ramp at the percent level; the Chebotarev races carry precisely the biases character theory dictates; and the Frobenius channel is statistically independent of the pair structure of the primes, as the simplicity of $\text{PSL}(2, 7)$ suggests it must be. The high conductor, elsewhere a computational burden, here acts as a microscope on the form factor.

None of the above bears on the Riemann Hypothesis for $L(s, \chi_8)$ or on the operator-theoretic open problem of [1]; GUE consistency is the expected behaviour of the universality class, and the conditional statements (race densities, cluster-expansion predictions) are labelled as such. Nothing here advances the twin prime conjecture; §5 establishes the opposite, a measured decoupling.

Data and code availability

All scripts, zero tables, Frobenius caches (regenerable; $\leq 10^9$ classes and $(10^9, 10^{10}]$ binned ψ_χ), figures, and per-run analysis notes with SHA-256 manifests are released with this paper; the certified zero data and the main program record are at [2]. The scripts, data, and per-run analysis notes are maintained at <https://github.com/selinaserephina-star/conductor21-companion> and archived with this deposit. Every number in this paper is produced by a named script on named data. Licences: the scripts under the MIT licence; the

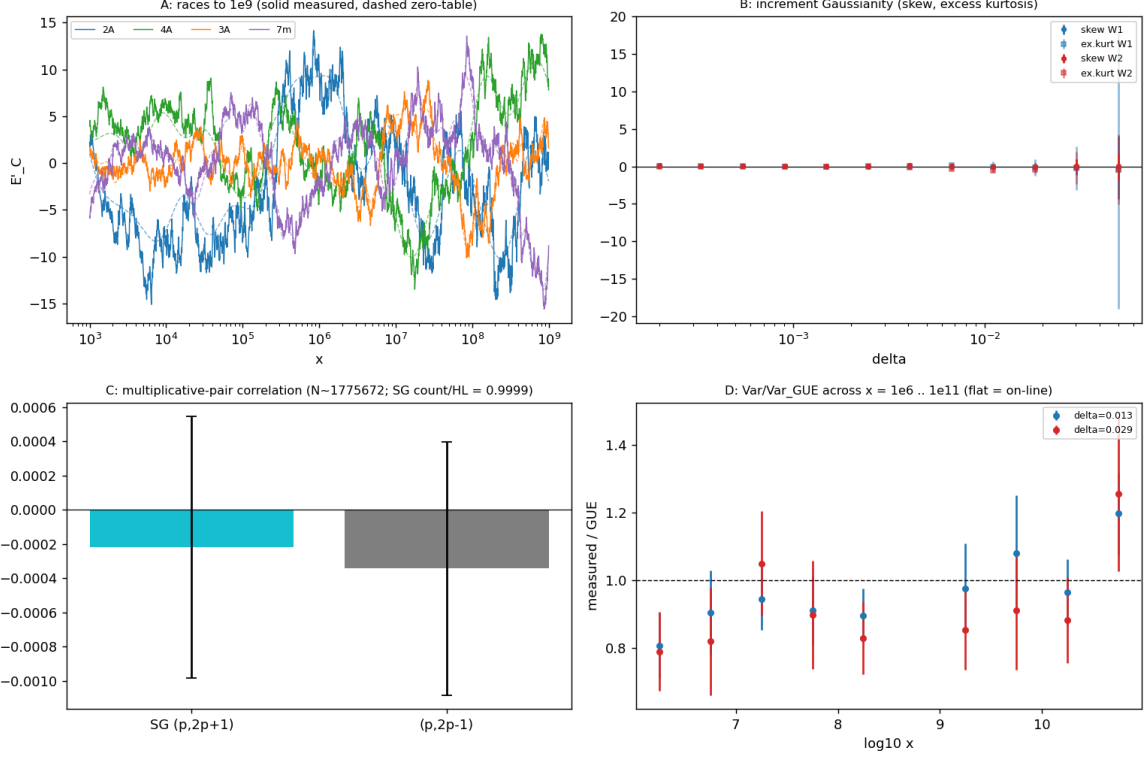


Figure 5: Closing measurements: races at 10^9 ; increment Gaussianity after exact shot noise; affine-pair independence with its Hardy–Littlewood count control; purity scaling across x .

paper, figures, and data tables under CC BY 4.0. Appendix B maps each section to its scripts, inputs, and outputs.

Verification note. Independent review of this work verified the theoretical inputs on which the statistics rest — the bias-mean formula and its square-root counts, the Frobenius–Schur indicators, and the framework identities, re-derived from first principles — and did not re-run the numerical computations; the scripts and data are released precisely so that any reader may perform that verification.

A Computational assistance

Following the convention of [1] (Appendix F): the computations of §§3,5–7 were designed and executed with the assistance of Claude (Fable 5, Anthropic), and those of §4 with Claude (Opus, Anthropic); the certified zero tables and engines are those of the main program. All mathematical claims were verified by computation; responsibility for the content rests with the authors.

B Scripts and data

Computations use Python 3 with NumPy [15], python-flint/FLINT [13], mpmath [16], and, for the $(10^9, 10^{10})$ sweep, Numba [17]; the companion zero tables were produced with PARI/GP [14]. Each script runs standalone against the released data seed and its own cached inputs; a SHA-256 manifest covers every released file, and the repository above carries the maintained copies.

script	sections	principal outputs
<code>bk_explore.py</code>	§3, §5 (twins)	<code>bk_offdiag_results.json</code> ; Fig. 1
<code>gapmap.py</code>	§5 (gaps), §8(i)	<code>gapmap_results.json</code> ; Fig. 2; class cache to 10^9
<code>chebrace.py</code>	§6	<code>chebrace_results.json</code> ; Fig. 3
<code>sivar.py</code>	§7	<code>sivar_results.json</code> ; Fig. 4
<code>finale.py</code>	§5 (affine), §6, §7, §8	<code>finale_results.json</code> ; Fig. 5
<code>sweep_1e10.py</code> , <code>retest_D.py</code>	§7 (scale purity), §8(iii)	<code>bins_1e9_1e10.npz</code> ; <code>retest_D_results.json</code>

Data. The certified zero table (153 zeros, $\gamma \leq 27.911$); the companion low-zero tables (χ_3 -pair, χ_6 , χ_7); exact Frobenius classes for all $p \leq 10^8$ (released; the $\leq 10^9$ cache and the $(10^9, 10^{10}]$ sweep regenerate from the included scripts); per-run result JSON files and figures. Provenance and per-run analysis notes accompany each script.

References

- [1] S. Stenberg and I. Balashov, *The Conductor-21 L-Function: Certified Zeros, Exact Laws, and a Density-Prime Operator*, Zenodo (2026), DOI 10.5281/zenodo.22669650.
- [2] S. Stenberg and I. Balashov, data deposit, Zenodo (2026), DOI 10.5281/zenodo.21958630.
- [3] W. Trinks, *Ein Beispiel eines Zahlkörpers mit der Galoisgruppe $PSL(3, 2)$ über $\mathbb{Q}$* , Manuskript, Universität Karlsruhe (1968). ($PSL(3, 2) \cong PSL(2, 7)$; cited as in Lubotzky et al., arXiv:2201.04103.)
- [4] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, Proc. Sympos. Pure Math. 24 (1973), 181–193.
- [5] D. A. Goldston and H. L. Montgomery, *Pair correlation of zeros and primes in short intervals*, in Analytic Number Theory and Diophantine Problems (Stillwater, OK, 1984), Progr. Math. 70, Birkhäuser Boston (1987), 183–203.
- [6] H. L. Montgomery and K. Soundararajan, *Primes in short intervals*, Comm. Math. Phys. 252 (2004), 589–617. DOI 10.1007/s00220-004-1222-4.
- [7] E. B. Bogomolny and J. P. Keating, *Random matrix theory and the Riemann zeros II: n -point correlations*, Nonlinearity 9 (1996), 911–935. DOI 10.1088/0951-7715/9/4/006.
- [8] M. Rubinstein and P. Sarnak, *Chebyshev’s bias*, Experiment. Math. 3 (1994), no. 3, 173–197. DOI 10.1080/10586458.1994.10504289.
- [9] N. Ng, *Limiting distributions and zeros of Artin L -functions*, Ph.D. thesis, University of British Columbia (2000).
- [10] N. M. Katz and P. Sarnak, *Zeroes of zeta functions and symmetry*, Bull. Amer. Math. Soc. (N.S.) 36 (1999), no. 1, 1–26. DOI 10.1090/S0273-0979-99-00766-1.
- [11] G. H. Hardy and J. E. Littlewood, *Some problems of ‘Partitio numerorum’; III: On the expression of a number as a sum of primes*, Acta Math. 44 (1923), 1–70. DOI 10.1007/BF02403921.
- [12] The LMFDB Collaboration, *The L -functions and Modular Forms Database*, <https://www.lmfdb.org>, object 8.16679880978201.21t14.b.a.

- [13] W. B. Hart, F. Johansson, et al., *FLINT: Fast Library for Number Theory*, <https://flintlib.org> (via python-flint).
- [14] The PARI Group, *PARI/GP*, Univ. Bordeaux, <https://pari.math.u-bordeaux.fr>.
- [15] C. R. Harris et al., *Array programming with NumPy*, Nature 585 (2020), 357–362. DOI 10.1038/s41586-020-2649-2.
- [16] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, <https://mpmath.org>.
- [17] S. K. Lam, A. Pitrou, and S. Seibert, *Numba: a LLVM-based Python JIT compiler*, Proc. Second Workshop on the LLVM Compiler Infrastructure in HPC (2015), 1–6. DOI 10.1145/2833157.2833162.